A ONE-WEEK MINI-PROJECT IN BIOINFORMATICS

Sequence Characteristics of Human TP53 and Gene-Expression Analysis in Breast Cancer (GSE42568)

Submitted by
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Project report

1. Introduction

Bioinformatics uses computational tools to study and interpret biological data. Large DNA sequences and high-throughput gene-expression datasets are difficult to analyse manually. In this project, Python, Linux command-line utilities and data-analysis libraries were used to read files, check data, calculate sequence properties and explore gene-expression patterns across sample groups.

2. Project Overview

This project contains two parts:

1. DNA sequence analysis (FASTA): The human TP53 nucleotide sequence was analysed to determine sequence length, individual nucleotide counts (A, T, G and C), and GC and AT percentages.

2. Gene-expression analysis: The breast cancer microarray dataset GSE42568 was explored to compare expression values between breast tumour samples and normal breast tissue samples.

3. Datasets and Tools Used

TP53 sequence: Human TP53 sequence from NCBI Gene, Gene ID 7157, in FASTA format.

Breast cancer expression matrix: GSE42568 series matrix from the NCBI Gene Expression Omnibus (GEO), containing 104 breast tumour samples and 17 normal tissue samples.

Tools and environment: Google Colab (Python 3), Linux command-line tools (wget, gunzip and head), Python file handling and built-in methods, pandas and matplotlib.pyplot.

4. Objectives

• Retrieve the human TP53 nucleotide sequence in FASTA format.
• Read and validate the sequence, count nucleotides and calculate GC and AT percentages.
• Inspect processed microarray data for GSE42568.
• Organise the expression matrix, separate tumour and control samples, and calculate group means and differences.
• Generate exploratory plots to compare expression patterns without making causal claims.

5. Methodology

5.1 FASTA sequence processing

The downloaded FASTA file (TP53.FASTA.txt) was read line by line using Python file handling. The header was identified from the line beginning with the “greater than” symbol. Sequence lines were stripped of whitespace and joined. The sequence was checked against the DNA alphabet A, T, G and C. Individual nucleotide counts were calculated, followed by GC and AT percentages.

GC% = [(G + C) / (A + T + G + C)] × 100
AT% = [(A + T) / (A + T + G + C)] × 100

5.2 Microarray data ingestion and quality checks

The GSE42568 series matrix text file was read into a pandas DataFrame, skipping metadata commentary lines beginning with “!”. The report records 54,675 probe sets across 121 biological samples, with zero missing values and zero duplicate rows.

5.3 Sample grouping and summary calculations

Sample descriptions from the metadata row !Sample_source_name_ch1 were used with accession IDs from !Sample_geo_accession to classify samples into 17 normal tissue controls and 104 breast cancer cases. Mean expression values were calculated for each group, and the tumour-minus-normal mean difference was determined for each probe.

5.4 Visualisation

The analysis used matplotlib to produce a bar chart comparing the top five probes, a histogram showing the distribution of expression measurements, and a box plot showing the spread of expression values for probe 201650_at across the two groups.

6. Results: TP53 Sequence Analysis

Metric	Calculated value
FASTA header	NC_000017.11:c7687490-7668421 TP53
Organism / Gene ID	Homo sapiens / 7157
Sequence length	38,121 bp
Adenine (A)	9,925
Thymine (T)	9,374
Guanine (G)	9,550
Cytosine (C)	9,272
GC percentage	49.37%
AT percentage	50.63%

7. Results: Gene-Expression Analysis (GSE42568)

The expression matrix contained 54,675 probe sets across 121 biological samples: 17 normal samples and 104 tumour samples. The report identified the following five probe sets with the greatest positive differences in mean expression (tumour mean minus normal mean).

Probe ID	Normal mean	Tumour mean	Difference
201650_at	6.29	11.74	+5.45
37892_at	3.60	8.98	+5.38

Probe ID	Normal mean	Tumour mean	Difference
232165_at	3.84	9.12	+5.28
201839_s_at	6.77	11.92	+5.16
232164_s_at	4.18	9.21	+5.04

8. Data Visualisation

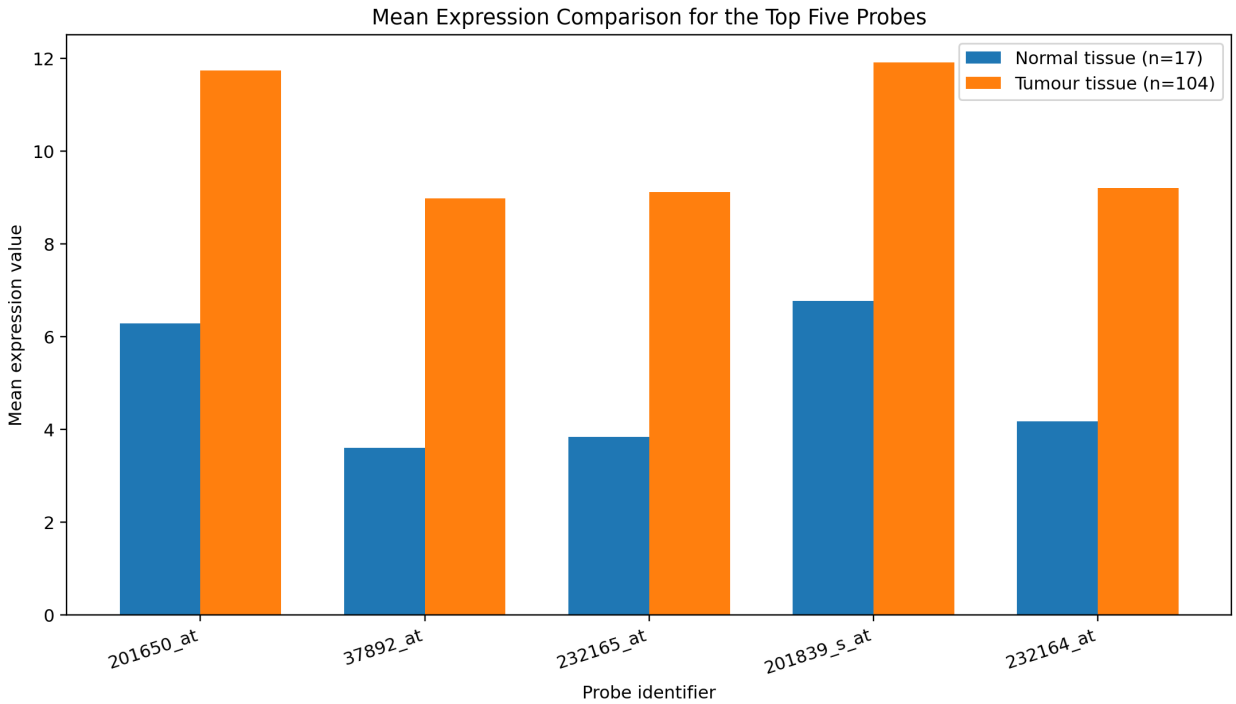


Figure 1. Mean expression values for the five probe identifiers listed in the submitted report. The bars are plotted from the normal and tumour group means reported in Table 2.

Other visualisations in the original analysis: The submitted report also includes a histogram of expression measurements (Figure 2) and a box plot for probe 201650_at comparing normal and tumour samples (Figure 3). These are not recreated here because the underlying plotted data were not supplied separately; the original Colab notebook is the source for those figures.

9. Interpretation and Limitations

The five listed probes had higher average expression values in the tumour group than in the normal group in this dataset. This is an exploratory observation and does not prove that the probes or their corresponding genes cause breast cancer. The sample groups are unequal in size (104 tumour samples versus 17 normal samples). Microarray findings require further biological validation, for example by RT-qPCR, before drawing stronger conclusions.

10. Conclusion

This project applied introductory bioinformatics workflows using Python, Linux file-inspection commands and pandas. The human TP53 sequence was checked and its nucleotide composition was calculated. The GSE42568 expression matrix was loaded, inspected and grouped into normal and tumour cohorts. Group-level expression differences were summarised and visualised. The work provided practical experience in sequence analysis, tabular biological data processing and exploratory visualisation.

11. Google Colab Notebook

The complete executable Python notebook and step-by-step outputs are available at:
Open the Google Colab notebook

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